Supplement to *Certified whole-domain bounds and equality barriers for the odd theta–Weil form*

Leonard van Hemert

13 October 2026 — Version 1.0.0

Abstract

This supplement supplies the analytic domain and tail arguments, finite matrix error formulas, and certificate correspondence for the main article. It also records the secondary signed-action and coupling results. All norms are in the original complex Hilbert space unless a coefficient norm is explicitly specified. Exact stored binary numbers define trial maps; floating solves and spectral decompositions select those numbers but do not establish inverses, rank, or positivity. The finite arithmetic certificates and their source files accompany this document.

Contents

S0 Foundations, domains, and primitive tails . . . 2
S0.1 Closed realization and domain-preserving kernels . . . 2
S0.2 Complete prime tail . . . 3
S0.3 Full gamma cutoff and complete cells . . . 3
S0.4 Theta and zero envelopes . . . 4
S1 Continuum features and the midpoint comparison . . . 5
S2 Dense capture and arithmetic spectral enclosures . . . 5
S3 Physical Gram and genuine rank . . . 6
S4 Fourier grids and the complete low-zero frame . . . 7
S4.1 Outward quadrature . . . 7
S4.2 Actual zeros, signs and multiplicity . . . 8
S4.3 The omitted zeros as a whole action bound . . . 8
S5 True masses and the weighted deficit . . . 8
S6 Whole-space midpoint error and frame action . . . 9
S7 Complete frame residual and center errors . . . 9
S8 The equality barrier and kernel-aware projection . . . 10
S9 Admissible enrichment and a constructive action estimate . . . 11
S10 The executed action certificate and its full Grams . . . 13
S11 Shifted residual theorem and the finite tradeoffs . . . 14
S12 Exact signed-action cancellation . . . 15
S13 A cached positive five-dimensional subspace . . . 15
S14 A fully coupled Schur criterion and its scope . . . 16

S0 Foundations, domains, and primitive tails

We use the definitions $\Phi, w, p, q, G, \mathcal{V}, A_W, E_8, J, J_8$ of Section 1 of the main article. Write $W = w^2$, $\Omega = \widehat{W}$, $\Xi = \widehat{\Phi}$, $Q = \|q\|^2$, $\tau = \|p\|^2$ and $S = I - q \otimes q/Q$. Pairings are sesquilinear, and $f \otimes f$ denotes the positive map $u \mapsto \langle u, f\rangle f$, with the corresponding consistent convention for adjoints. The whole-line normalization equals $2\,dx$ on the positive half-line for products of odd functions. In particular the positive-frequency measure is $r_8(t)\,dt/\pi$, not $r_8(t)\,dt/(2\pi)$.

S0.1 Closed realization and domain-preserving kernels

The original signed realization and unconditional pole-null jet identity are [1, Lemma 5.1]; its closed form and compact core are proved in [1, Lemma 2.2]. We spell out the additional facts needed for this article. The positive symbol has the absolutely convergent representation

$$b(t) = \sum_{j\ge 0} \frac{(t/2)^2}{(j+1/4)((j+1/4)^2 + (t/2)^2)}.$$

Multiplication by bounded w followed by the closed multiplier $b(D)^{1/2}$ is closed: convergence of its input and output implies convergence of wu and its multiplier image, so closedness of the multiplier applies. Its squared norm defines the closed form g. The prime majorant below and the finite-rank pole show that the remaining signed form is a bounded selfadjoint perturbation. Thus A_W is selfadjoint on $\mathrm{Dom}(G)$ and has form domain $\mathcal{V}$.

Writing $a_j = 2j + 1/2$ gives

$$b'(t) = 4t \sum_{j\ge 0} \frac{a_j}{(a_j^2 + t^2)^2}, \qquad t > 0.$$

The step-two unimodal sum is bounded by half its integral plus twice its supremum. Consequently $b'(t) \le 9/t$ for $t \ge 1$, while $b(1) \le \frac{1}{4}\sum (j+1/4)^{-3} < 17$. Hence

$$0 \le b(t) \le 17 + 9\log(1 + |t|). \tag{S0.1}$$

For each fixed frequency, $W\sin(tx)$, $W\sinh(x/2)$ and every fixed derivative of Φ are Schwartz. The entire theta series and its differentiated series converge with a common superexponential majorant on the real line. Equation (S0.1) therefore places the corresponding physical columns in $\mathrm{Dom}(G)$.

The score $q = \Phi'/w$ and first pole-null jet $h = (\Phi''' - \Phi'/4)/w$ also belong to $\mathrm{Dom}(G)$ and to $\ker A_W$. Here is the full domain argument underlying that statement. For every fixed j, termwise differentiation gives $|\Phi^{(j)}(x)| \le D_j e^{2j|x|}\Phi(x)$ with finite D_j: a polynomial in $y = \pi n^2 e^{2|x|}$ multiplies each theta atom, and the residual sum is bounded by $\sum n^{4+2j} e^{-\pi(n^2-1)} < \infty$. Division by the positive first theta atom bounds the ratios and shows that q, h are square integrable with all required theta-weighted derivatives. Their weighted transforms are

$$\widehat{wq}(z) = -iz\Xi(z), \qquad \widehat{wh}(z) = iz(z^2 + 1/4)\Xi(z).$$

They vanish at every actual nontrivial zero, without assuming its ordinate real. The second also vanishes at the poles $z = \pm i/2$. Apply the signed explicit formula first to compactly cut off physical tests and then remove the cutoffs. The relative-theta bounds just given control every differentiated cutoff error; contour shifts within $|\Im x| < \pi/4$ and the zero-count bound below dominate the paired zero sum. The full geometric side contains the contact, every prime power and both poles before odd restriction. Thus its mixed pairing with either vector vanishes on the compact form core. By

continuity in the form norm, it vanishes on $\mathcal{V}$. The representation theorem gives $A_W q = A_W h = 0$ and their operator-domain membership. For the score, the nonzero pole values are included in this signed identity; they are not discarded as if the score were pole-null.

More generally, let K be a closed subspace of $\ker A_W$. If $A_W = G - D$ with D bounded, then $GP_K = DP_K$ is bounded. It follows that P_K preserves both $\mathrm{Dom}(G)$ and $\mathcal{V}$ and

$$\mathfrak{a}(u) = \mathfrak{a}((I - P_K)u), \qquad u \in \mathcal{V}. \tag{S0.2}$$

Indeed the spectral projection onto the kernel reduces A_W; any subprojection inside that zero block also reduces it. Alternatively expand the form and use its vanishing mixed pairings. Since $|d_{8,+} - b| \le |c_\Gamma| + \|k_8\|_\infty$, the operators E_8 and G differ by a bounded selfadjoint operator. They have identical operator and form domains. The positive compressed operator $\mathcal{E}$ on $q^\perp$ has domain $\mathrm{Dom}(G) \cap q^\perp$ and value $SE_8 u$: the missing pairing with q is bounded because $E_8 q \in H$. This justifies every resolvent and projection used in the main article.

S0.2 Complete prime tail

Put $q_0 = e^{-\pi}$, $d = 1+16q_0^3/(1-6q_0)$ and $a = 3/(2\pi)$. The inequalities $j^4 \le 16\,6^{j-2}$ and $j^2-1 \ge j+1$ for $j \ge 2$ retain every theta atom and give

$$\Phi(x) \le 4\pi^2 e^{9|x|/2-\pi e^{2|x|}}(d - ae^{-2|x|}).$$

For a shift $\log n$, write $A = e^{2|x|}$, $B = e^{2|x-\log n|}$, $t = \sqrt{AB} \ge n$. Using $A + B \ge 2t$ and $(d - a/A)(d - a/B) \le (d - a/t)^2$, and dropping only the positive cosh denominator, gives

$$w(x)w(x - \log n) \le 4\pi^2 t^{9/4} e^{-\pi t}(d - a/t).$$

This decreases for $t \ge 2$, since its logarithmic derivative is at most $9/8 + 1/6 - \pi < 0$. Therefore

$$\|J - J_8\| \le 8\pi^2 \sum_{n \ge 9} \Lambda(n) n^{7/4} e^{-\pi n}(d - a/n) < t_8 := \frac{32237}{10^{12}}. \tag{S0.3}$$

For an exact rational verification take $\pi_+ = 355/113$, $q_+ = 2160875733/(5 \cdot 10^{10})$ and $\delta_+ = 348730539/(2 \cdot 10^{11})$. Then $\pi < \pi_+$, $e^{-\pi} < q_+$ and $d \le 1 + \delta_+$. Since $\Lambda(n)n^{7/4} \le n^3$, the last sum is at most

$$8\pi_+^2(1 + \delta_+)\,\frac{9^3 q_+^9}{1 - q_+(10/9)^3} < \frac{32237}{10^{12}}.$$

The rational inequalities follow by cross multiplication; the elementary π and exponential endpoints follow from Machin's formula with alternating remainders and the exponential Taylor remainder. The same summable majorant proves operator-norm convergence before the split. Thus this is a whole operator bound, not a bound on a sampled prime sum.

S0.3 Full gamma cutoff and complete cells

The exact principal-log integral for ψ [2, 5.9.13] yields, for $|t| \ge 2$,

$$m_\Gamma(t) \ge \log(|t|/(2\pi)) - 211/300.$$

To see the explicit error, $\coth s \le 1/s + s/3$ follows by comparing Taylor coefficients in $(3 + s^2)\sinh s - 3s\cosh s$. Hence $1/(1 - e^{-x}) - 1/x \le 1/2 + x/12$. The integral at $5/4 + it/2$ bounds

$|\psi(z) - \log z|$ by $34/75$; recurrence back to $1/4 + it/2$ costs at most $1/t^2 \le 1/4$. The six actual prime powers have $\|k_8\|_\infty < 13/2$. Since $\log(10000/(2\pi)) > 29/4$,

$$d_8(t) > 7/150 \quad (|t| \ge 10000), \qquad r_8(t) = 0 \quad (|t| \ge 10000). \tag{S0.4}$$

For reproducible exact checks, the square-root lower endpoints for $n = 2, 3, 4, 5, 7, 8$ are respectively $7/5, 5/3, 2, 11/5, 13/5, 14/5$. Logarithm upper endpoints may be checked by positive exponential Taylor sums. The logarithm inequality uses $e < 11/4$, $e^{1/4} \le 4/3$ and $\pi < 22/7$.

On $[0, 10000]$, all 80000 closed cells of width $1/8$ are covered. Outward evaluation classifies 79524 as zero and 476 as possibly positive. The latter are refined to width $1/256$; their 11476 possibly nonzero subcells lie in 2905 cells of width $1/64$. For a subcell with midpoint c, halfwidth h and lower endpoint $a > 0$, the full gamma derivative bound is

$$L(a) = \min(10, 2/a^2 + \pi/(2a)), \qquad L(0) = 10.$$

It follows by differentiating the entire digamma series and comparing the remaining positive tail with its integral. Thus the center $\Re\psi(1/4 + ic/2) - \log\pi$ plus $[-Lh, Lh]$ encloses the full symbol. Evaluating all six cosines on the whole interval and taking the positive part gives upper and lower rectangles for r_8. Every upper mass is rounded upward to a dyadic endpoint of denominator 2^{80}, then to the exact upward binary64 endpoint used to define B. Their sum is below 14. Zero cells, refined cells and (S0.4) cover the entire frequency line by evenness.

S0.4 Theta and zero envelopes

For $x \ge 0$, the full lattice gives $\Phi(x) \le 80e^{9x/2 - 3e^{2x}}$ and $W(x) \le 160e^{4x - 3e^{2x}}$. Contour shifting gives

$$|\Omega(t)|, |v(t)| \le 5e^{-|t|/4}, \qquad |t\Xi(t)| \le 5|t|e^{-|t|/4}, \tag{S0.5}$$

where $v(t) = 2\int_0^\infty W(x)\sinh(x/2)\sin(tx)\,dx$. For completeness, on $|\Im x| \le 1/4$ use $\pi\cos(1/2) > 21/8$ and $|\cosh((x + i/4)/2)| \ge (127/128)\cosh(x/2)$. With $q = 1/13$ and

$$S_0 \le q + q^4/(1 - q^3), \quad S_2 \le q + 4q^4/(1 - 3q^3), \quad S_4 \le q + 16q^4/(1 - 6q^3),$$

the complete strip bounds are $|W(z)| \le (256/127)(40S_4 + 19S_2) < 10$ and

$$\int |\Phi(x + i/4)|\,dx, \quad \int |W(x + i/4)\sinh((x + i/4)/2)|\,dx \le (128/127)(15S_2 + 25S_0) < 5.$$

The analogous W integral is smaller. They follow by $t = e^{2|x|}$ and integration of the positive exponential-polynomial majorants. Vertical contour ends vanish superexponentially, proving (S0.5). These estimates also give the looser weighted $1/8$-strip integrals bounded by 120 used below.

The unconditional zero count, with both signs and multiplicity, is bounded for $L \ge e^2$ by $2L\log L$. The explicit bound of [3, Corollary 1.2] is $|N_+(T) - \frac{T}{2\pi}\log(T/(2\pi e))| \le 0.1038\log T + 0.2573\log\log T + 9.3675$ for $T \ge e$. Twice this upper bound is at most $2T\log T$ for $T \ge e^2$: use $\log\log T \le \log T$, bound the main coefficient by $1/\pi$, and check the endpoint; the difference thereafter increases. This retains multiplicity and both signs. Stieltjes integration then gives the complete tail

$$Z_a(L) = e^{-aL}\left[2L\log L + \frac{2(\log L + 1)}{a} + \frac{4}{a^2 L}\right], \qquad a > 0. \tag{S0.6}$$

Indeed integrate the counting majorant against ae^{-at} on $[L, \infty)$, twice integrate by parts, and use $\int_L^\infty e^{-at}/t\,dt \le e^{-aL}/(aL)$. The bound includes possible off-line zeros: if $\rho = 1/2 + i\gamma$, then $|\Im\gamma| < 1/2$. Cauchy–Schwarz and the complete theta envelope give $|\widehat{wu}(\gamma)| \le 11\|u\|$. Thus exponentially decaying samples may be paired against every Hilbert test in the full signed zero formula, rather than only other trial vectors.

S1 Continuum features and the midpoint comparison

Put $f_t = S[w\sin(tx)]$ and $p_Y = Sp$. On the common coefficient space $L^2(r_8(t)\,dt/\pi) \oplus \mathbb{C}$, the continuum feature map has positive product

$$R_q = SR_8S = \int_0^{10000} f_t \otimes f_t\, r_8(t)\,dt/\pi + 2p_Y \otimes p_Y.$$

Let $I_i = [k_i/64, (k_i+1)/64]$, $t_i = (2k_i+1)/128$ and let μ_i^+ be the exact stored upward masses. Define $Be_i = \sqrt{\mu_i^+}\, f_{t_i}$, $Be_p = \sqrt{2}p_Y$ and $R_m = BB^*$. The square roots are mathematical square roots of exact dyadics.

Because $|x| \le 2|\sinh(x/2)|$ and $\tau < 3/500$,

$$\|f_t - f_{t_i}\|^2 \le |t-t_i|^2 \int x^2 W(x)\,dx \le 4\tau|t-t_i|^2.$$

The full mass is below 14 and $|t-t_i| \le 1/128$, whence

$$\int \|f_t - f_{t_i}\|^2\, r_8(t)\,dt/\pi \le \delta_0 := 21/1024000. \tag{S1.1}$$

For any $a > 0$, vector Young's inequality on every whole cell proves

$$R_q \le (1+a)R_m + (1+1/a)\delta_0 I. \tag{S1.2}$$

Replacing the true cell mass by its upper mass is legitimate only on the nonnegative midpoint term. The pole is exact in both maps; enlarging it by $1+a$ is conservative.

Let $G_0(i,j) = [\Omega(t_i - t_j) - \Omega(t_i + t_j)]/2$, $G_0(p,i) = v(t_i)$, $G_0(p,p) = \tau$, $b_i = -t_i\Xi(t_i)$ and $b_p = -1/4$. For $D_\mu = \operatorname{diag}(\sqrt{\mu_i^+}, \sqrt{2})$,

$$B^*B = D_\mu(G_0 - bb^*/Q)D_\mu \le \Gamma,$$

where Γ replaces Q by 12 and the pole norm by $3/500$. The difference is positive semidefinite, since $Q < 12$ and $\tau < 3/500$. All difference, sum, score and pole entries remain.

S2 Dense capture and arithmetic spectral enclosures

The exact dyadic 2906×56 matrix Z and positive diagonal Λ satisfy the certified full dense residual bound

$$\|\Gamma - Z\Lambda Z^*\| < 3/12500.$$

Let P be the actual coefficient projection onto $\operatorname{ran} Z$. Since $(I-P)Z = 0$ and $\Gamma \ge 0$, $0 \le B(I-P)B^* \le (3/12500)I$. Once $N = Z^*Z > 0$ is established, $P = ZN^{-1}Z^*$. Put $C = BZ$ and $R_{56} = (11/10)CN^{-1}C^*$. Using (S1.2) with $a = 1/10$ and the complete prime tail gives

$$\mathfrak{a}(z) \ge E_8(z) - \langle R_{56}z, z\rangle - \rho\|z\|^2, \quad \rho = t_8 + 11\delta_0 + \frac{11}{10}\frac{3}{12500} = \frac{979236349}{2000000000000}. \tag{S2.1}$$

This holds throughout $\mathcal{V} \cap q^\perp$. No physical mixed block is removed by the coefficient-space decomposition.

The center entry error for Γ is below $3.124405 \cdot 10^{-9}$ in operator norm. Near Fourier arguments use outward dyadic endpoint pairs; far arguments use (S0.5). Each short scalar entry

expression receives 10^{-12} for endpoint conversion, products, square-root scales and symmetrization. Absolute row sums receive the positive summation guard below. The full center Frobenius residual is 0.0002391007522 to the displayed precision; its guarded upper endpoint is below $0.000239106 < 3/12500$. The entire dense residual is evaluated, not just retained columns.

Here are the common arithmetic rules used throughout the certificate. For binary64 unit roundoff $u = 2^{-53}$, write $\gamma_n = nu/(1-nu)$. A length-m matrix dot product receives γ_{2m+10} times its absolute-product sum. Positive sums used to bound those absolute products are themselves enlarged by their corresponding γ factor. A Frobenius norm receives a serial square/sum bound for all entries, followed by an outward square root. Symmetrization and short final scalar formulas receive their stated additional absolute allowance. Nonnegative row sums are divided by $1-\gamma_n$ and rounded upward. Every intermediate is finite and away from overflow; outward nextafter steps cover conversion underflow. No unbounded cancellation is hidden in a relative sum error.

For a real symmetric center A, let the exact dyadic locator V, D have orthogonality defect $o = \|V^*V - I\| < 1$, full residual $r = \|AV - VD\|$ and $L = \|D\|$. Its exact polar unitary $U = V(V^*V)^{-1/2}$ obeys

$$\|U^*AU - D\| \le \frac{r}{\sqrt{1-o}} + 2L\sqrt{1+o}\left(\frac{1}{\sqrt{1-o}} - 1\right). \tag{S2.2}$$

Expand $AU - UD$ into $(AV - VD)(V^*V)^{-1/2}$ and the remaining commutator. Functional calculus bounds the latter by twice L times the distance of $(V^*V)^{-1/2}$ from I. Add any whole center error $\|A_{\mathrm{true}} - A\|$ before using the spectral endpoint. This proves that eigensolvers supply locators only. The later small comparisons use exact-input 192-bit Arb products and Frobenius bounds in (S2.2).

S3 Physical Gram and genuine rank

The coefficient and physical metrics are different:

$$N = Z^*Z, \qquad H_C = C^*C = Z^*B^*BZ.$$

The full coefficient defect is below $3.627 \cdot 10^{-11}$ after dot and row guards. Hence

$$(1 - 10^{-8})I \le N \le (1 + 10^{-8})I. \tag{S3.1}$$

The physical Gram uses the actual Q and τ, not their loose upper substitutes. The complete norm integrals are

$$Q = 2\int_0^\infty \frac{\Phi'^2 \cosh(x/2)}{\Phi}\,dx, \qquad \tau = 2\int_0^\infty \frac{\Phi \sinh^2(x/2)}{\cosh(x/2)}\,dx.$$

Put $t = e^{2x}$ and $y_n = \pi n^2 t$. The first two atom polynomials are $4y^2 - 6y$ and $-8y^3 + 30y^2 - 15y$. For $y \ge 3$, positivity of the first and the complete lattice sum imply $|\Phi'| \le 80t\Phi$. Consequently both doubled integral tails beyond $x = 2$ are bounded by

$$512000\int_{54}^\infty t^4 e^{-3t}\,dt < 2^{-180}.$$

On $[0, 2]$ the atoms $n \ge 9$ satisfy $\sum_{n\ge 9} n^r e^{-\pi t(n^2-1)} \le 2\,9^r e^{-240}$ for $r = 4, 6$. Relative to the retained positive eight-atom sum, the omitted function and derivative are each below 2^{-300}. Expanding the quotient shows that their total integral effect is below 2^{-250}. The common allowance is therefore $2^{-250} + 2^{-180}$.

Native meromorphic Arb integration on $[0,2]$ encloses the finite integrals, with absolute and relative tolerance 2^{-100}. A possible zero in a denominator ball produces a nonfinite ball rather than an analyticity assertion. The finite returned balls, enlarged by both complete allowances, give

$$Q = 11.067665747142874946319167822 9 \ \pm 6.97 \cdot 10^{-29},$$
$$\tau = 0.00571290852978025013036375087 \ \pm 4.55 \cdot 10^{-30}.$$

The actual whole Gram error is below $3.138 \cdot 10^{-9}$. Compression adds less than $5.785 \cdot 10^{-11}$, and multiplying the input error by $\|Z\|^2 \le 1 + 10^{-8}$ gives total below $3.2 \cdot 10^{-9}$. Score products of either sign use both Q endpoints. The exact pole entry is $2(\tau - 1/(16Q))$ and the cross term is $v(t_i) + b_i/(4Q)$ before mass scaling.

For the saved physical spectral locator, safe rational upper errors are $d = 32/10^{10}$, $o = 43/10^{12}$ and $r = 115/10^{13}$ in (S2.2). They prove

$$H_C \ge \frac{21}{10^8} N, \qquad H_C < \frac{32}{125} I. \tag{S3.2}$$

It follows that the actual physical rank is 56 and the coefficient preimage cost is at most $\kappa_B \le \sqrt{10^8/21} < 2182.178903$. No enlarged enriched-trial inverse or rank is inferred from this.

S4 Fourier grids and the complete low-zero frame

S4.1 Outward quadrature

The basic grid evaluates Ω, Ω' at $j/64$ through 100 and $v, -t\Xi(t)$ at the half-step frequencies. Native 192-bit Arb Legendre roots and weights of order 256 are mapped to $[0,3]$. The Bernstein ellipse of parameter $r = 23/20$ has imaginary part less than $1/4$, real part below $16/5$ and $|z| < 4$. The strip estimates in S0 give the common integrand bound $M = 240 \cdot 3^{25}$. Chebyshev coefficients satisfy $|a_j| \le 2Mr^{-j}$; positivity and polynomial exactness through degree 511 give the doubled integral error

$$E_{256} = 24Mr^{-512}/(1 - r^{-1}) + 32\,2^{-128} + 2^{-1160}.$$

The second term covers all omitted theta atoms; the last covers the whole physical tail. For example $t = e^{2x}$ reduces that tail to $160e^{-1200}(729^2/3 + 2 \cdot 729/9 + 2/27) < 2^{-1175}$. Every transform ball is enlarged before conversion to outward integer endpoints over 2^{80}. For the score multiply the enclosed $\Xi(t)$ by the exact frequency and retain its enlarged radius.

Arbitrary zero arguments use an order-20 Taylor grid at $j/8$ through 128, with native quadrature order 320. Derivative j costs 4^j in the ellipse bound; its complete base error is

$$E_{320} = 24 \cdot 240 \cdot 3^{32} r^{-640}/(1 - r^{-1}) + 32\,2^{-128} + 2^{-1100}.$$

For $j \le 19$, the physical derivative tails are bounded by $160 \int_{400}^{\infty} t^6 e^{-3t}\,dt < 2^{-1100}$. At displacement $|h| \le 1/16$ the entire Taylor remainder is at most $120(2|h|)^{20}$, using the exponential moment bound from S0. Horner products, coefficient radii, frequency addition and native zero location errors are priced. Global derivative bounds are $|\Omega'| < 11/100$ and $|v'| < 3/250$. Far arguments retain (S0.5) with a downward dyadic argument reserve; both sum and difference arguments are included through 6000.

S4.2 Actual zeros, signs and multiplicity

Native FLINT 3.6.0 `zeta_zeros` and `zeta_nzeros` [4] certify 2469 disjoint actual positive roots on the critical line and the full multiplicity count $N(3000) = 2469$. Their last upper ordinate is below 3000; hence the roots exhaust the count and are simple within this finite interval. The intervals are stored outward over 2^{128}. Both signs are retained by oddness. Define $V_0 e_j = w \sin(\gamma_j x)$, $A_0 = V_0^* C$ and $G_z = V_0^* V_0$. At actual zeros the score transform vanishes exactly. Thus

$$S_{3000} = 2V_0 V_0^*, \quad L = C^* S_{3000} C = 2A_0^* A_0, \quad O = C^* S_{3000}^2 C = 4A_0^* G_z A_0.$$

The complete sample entries use $\sqrt{\mu_i^+}[\Omega(\gamma - t_i) - \Omega(\gamma + t_i)]/2$ and pole entry $\sqrt{2}v(\gamma)$. All 2469^2 zero correlations and all $2469 \cdot 2906$ zero/feature correlations are included. The full certified errors are

$$\|\Delta A_0\|_F \le 2 \cdot 10^{-12}, \quad \|\Delta G_z\| \le 3.2 \cdot 10^{-11}, \quad \|\Delta L\| \le 10^{-11}, \quad \|\Delta O\| \le 3.1 \cdot 10^{-10}.$$

To certify a small positive gap, the rectangular factor A_0 is checked directly. Its singular-factor locator has left and right orthogonality defects at most $10^{-11}, 2 \cdot 10^{-13}$ and full residual at most $3 \cdot 10^{-14}$. Subtracting this residual and the complete sample error from $\sigma_{\min}\sqrt{1 - o_L}\sqrt{1 - o_R}$ gives at least $3.2586 \cdot 10^{-8}$. Consequently $L \ge 10^{-15} I$. This argument does not subtract the larger additive L-error from an unverified tiny eigenvalue.

S4.3 The omitted zeros as a whole action bound

The actual largest midpoint is $T = 313317/128$. If $f = B\xi$ with minimal coefficient preimage, $\|\xi\| \le \kappa_B \|f\|$. The sine sum on the 1/8 strip has size at most $4e^{T/8}\|\xi\|$, and the pole coefficient at most $2\|\xi\|$. Using the full strip integrals bounded by 120 yields, at every actual zero,

$$|\widehat{wf}(\gamma)| \le 1000 \kappa_B e^{(T - |\Re\gamma|)/8} \|f\|.$$

The subtracted score vanishes at these zeros. Squaring this bound and using (S0.6) bounds the form remainder; pairing it with the whole Hilbert evaluation bound $11\|v\|$ bounds the operator remainder:

$$e_L = 10^6 \kappa_B^2 e^{T/4} Z_{1/4}(L), \qquad d_L = 11000 \kappa_B e^{T/8} Z_{1/8}(L).$$

At $L = 3000$ these satisfy $e_L < 3 \cdot 10^{-43}$ and $d_L < 1.4 \cdot 10^{-18}$. All multiplicities, both signs and possible high off-line zeros remain. The mixed formula extends from compact core tests by the same theta domination as S0, proving $\|(A_W - S_{3000})|_{\operatorname{ran} C}\| < d_L$. Together with (S3.2) and $L \ge 10^{-15} I$ this gives $\mathfrak{a}|_{\operatorname{ran} C} \ge (38/10^{16}) I$. It supplies no positive lower bound for the infinite complement.

S5 True masses and the weighted deficit

For each subcell, the lower rectangle from S0 gives an outward lower mass ℓ_i on the cell I_i, rounded downward over 2^{80}. Thus $0 \le \ell_i \le \mu_i := \int_{I_i} r_8(t)\, dt/\pi \le \mu_i^+$. The exact deficits $d_i = \mu_i^+ - \ell_i$ have sum below 0.29826603. For the true-mass midpoint operator R_t, including the identical pole,

$$0 \le R_m - R_t \le R_d := \sum_i d_i f_{t_i} \otimes f_{t_i}.$$

A trace estimate is valid but unnecessarily large. Instead every entry of the actual score Gram is bounded by the positive upper Gram $\overline{G} = G_0 - bb^*/12$. Its full entry intervals use the near Fourier endpoints and all far envelopes. If $A_{ij} \geq |\overline{G}_{ij}|$ and $w_i > 0$, the weighted Schur estimate is

$$\|R_d\| \leq \max_i \frac{\sum_j A_{ij} d_j w_j}{w_i}.$$

The weights are exact positive integer multiples of 2^{-30}, bounded below by $1/1024$. Their method of selection imposes no convergence premise. Entry errors receive 10^{-12}, and the positive row products, sums and division receive $\gamma_{2\cdot 2905+20}$, an outward step and a final exact 10^{-15} allowance. The guarded maximum is below 0.008552658376, giving

$$\|R_m - R_t\| < 427633/50000000 = 0.00855266. \tag{S5.1}$$

The pole deficit is zero; this removes no physical pairing with a pole test vector because the remaining feature map is kept in full.

S6 Whole-space midpoint error and frame action

Regard the continuum and true-mass midpoint feature maps as maps on the same L^2 input space. By (S1.1) their difference has norm at most $\sqrt{\delta_0}$. Expanding their positive products gives

$$\|R_q - R_t\| \leq 2\sqrt{\|R_t\|\delta_0} + \delta_0.$$

The dense capture and exact factor provide $\|R_t\| \leq \|R_m\| \leq (1 + 10^{-8}) \max \Lambda + 3/12500 < 0.255985513$. Thus the whole-space bound is

$$\|R_q - R_m\| \leq \epsilon_R := 2\sqrt{0.255985513\delta_0} + \delta_0 + 0.00855266. \tag{S6.1}$$

On the score quotient, the exact domain identity is $\mathcal{E} = A_W + R_q + S(J - J_8)S$. Therefore on $\operatorname{ran} C$,

$$\|\mathcal{E} - (S_{3000} + R_m)\| \leq \epsilon_R + t_8 + d_{3000} < 0.013155646076930 =: \eta.$$

This last action bound is restricted to $\operatorname{ran} C$; it is never applied to newly enriched zero directions. The whole-space bound (S6.1), by contrast, is independent of a trial space.

S7 Complete frame residual and center errors

This section supplies the finite realization of main Lemma 2.1. For $s > 0$, let $Y_0 = CX$ with exact dyadic X. Put $G_B = B^*B$, $Q_b = V_0^*B$ and $G_z = V_0^*V_0$. Then, with $E_m = S_{3000} + R_m$,

$$r = C - (E_m + s)Y_0 = BD_1 + V_0D_2, \quad D_1 = Z(I - sX) - G_BZX, \quad D_2 = -2A_0X,$$

$$r^*r = D_1^*G_BD_1 + D_1^*Q_b^*D_2 + D_2^*Q_bD_1 + D_2^*G_zD_2.$$

Every term is a full physical Gram. In particular R_mC is not replaced by CH_C, and the residual is not projected into $\operatorname{ran} C$. The full Q_b Frobenius error is below $7.215 \cdot 10^{-13}$; all other input errors are those of S3–S4.

If binary coefficient maps differ from the exact D_1, D_2 by e_1, e_2, their physical map error is at most $e_d = \sqrt{\|G_B\|}e_1 + \sqrt{\|G_z\|}e_2$. Let S_c approximate the Gram of the binary map with error r_e. Then the exact residual Gram has error bounded above by

$$s_e = r_e + 2\sqrt{\lambda_{\max}(S_c) + r_e}\, e_d + e_d^2.$$

This prices input maps before the matrix Young comparison. If h_e, l_e, m_e bound respectively the physical Gram, zero compression and midpoint compression errors, the remaining exact-center errors obey

$$\begin{aligned} q_e &\le h_e\|X\|^2 + \|Q_c - X^*H_cX\|_F, \\ k_e &\le 2h_e\|X\| + (l_e + m_e + sh_e)\|X\|^2 \\ &\qquad + \|K_c - (H_cX + X^*H_c - X^*(L_c + M_c + sH_c)X)\|_F. \end{aligned}$$

The displayed center differences are computed using exact binary inputs and exact rational s in Arb. Thus shift conversion, all inner and outer products and symmetrization are included. Each raw numerical guard is enlarged by 10^{-12}; a further 10^{-6} is added to the final comparison price.

The complete upper prices, with $s = \mu - \rho$, are

μ	Young parameter	complete price
1/20	2/3	100157725018/125000000000
1/25	1/2	450253581783/500000000000

They are strictly below the respective targets

$$\frac{400631}{500000}(1 - 10^{-8}), \qquad \frac{225127}{250000}(1 - 10^{-8}),$$

with exact margins $9592169/(5 \cdot 10^{13})$ and $20685723/(25 \cdot 10^{12})$. Factorization in main Lemma 2.1 and (S2.1) give the stated frame-only comparisons. Their role is to exhibit the effect of keeping matrix correlation and to supply the residual directions used next.

S8 The equality barrier and kernel-aware projection

This supplies the scalar certificate for main Theorem 5.1. With $h = (\Phi''' - \Phi'/4)/w$, set $a = \langle h, q\rangle/Q$, $k = h - aq$ and $D = \|k\|^2$. S0 proves $k \in \mathrm{Dom}(G) \cap \ker A_W \cap q^\perp$. Its transform $it(t^2 + 1/4 + a)\Xi(t)$ is not identically zero, so $D > 0$. The degree-five atom polynomial for wh is

$$-32y^5 + 360y^4 - 1056y^3 + 810y^2 - 90y.$$

For $y \ge 3$ its absolute value is at most $301y^5$. The full lattice comparison $\sum n^{10}q^{n^2-1} \le 1 + 1024q^3/(1 - 58q^5) < 2$, $q < 1/20$, gives $|wh|/\Phi < 20000e^{6x}$. Both doubled tails beyond $x = 2$ of the norm and score pairing are bounded by

$$10^{11}\int_{54}^{\infty} t^8 e^{-3t}\, dt < 2^{-140}.$$

On $[0, 2]$, the omitted degree-ten lattice satisfies $\sum_{n\ge 9} n^{10}e^{-\pi t(n^2-1)} \le 2\,9^{10}e^{-240}$. The three relative errors for Φ, Φ', wh are below 2^{-270}. Expanding the two quotient integrands bounds their integral

errors by $12\,2^{85}2^{-270} < 2^{-170}$. The common allowance 2^{-130} is added to each meromorphic Arb integral before projection. This gives

$$\langle h, q\rangle = 73.9389206536588932761816 \ \pm 6.26\cdot 10^{-23},$$

$$\|h\|^2 = 54152.6907182449014086321392 \ \pm 3.28\cdot 10^{-23}.$$

In particular $a > 6$ and $D < 55000$.

Every term of R_m is nonnegative. The 635 stored cells with $t_i \le 20$ give the valid lower sum

$$\langle R_m k, k\rangle \ge \sum_{t_i\le 20} \mu_i^+ t_i^2 (t_i^2+25/4)^2 \Xi(t_i)^2 > 1875.$$

The exact dyadic masses and native complex Ξ evaluations yield

$$1876.739373063928194740237244735 2786$$

with radius below $2\cdot 10^{-31}$. Thus $r = \langle R_m k, k\rangle/D > 3/88$. The common feature-map triangle inequality gives $\langle R_q k, k\rangle \le (\sqrt{\langle R_m k, k\rangle} + \sqrt{\delta_0}\|k\|)^2$. Using $\mathfrak{a}(k) = 0$, the dense capture and full prime tail proves

$$\begin{aligned}\frac{\langle R_{56}k, k\rangle - E_8(k)}{D} &\ge f(r),\\ f(r) &:= r/10 - 2\sqrt{\delta_0 r} - \delta_0\\ &\quad - (11/10)(3/12500) - t_8.\end{aligned}$$

The derivative is positive for $r \ge 3/100$, and exact outward arithmetic gives $f(3/88) > 7/5000$. This is the strict main barrier. Testing $R_{56} \le \theta(\mathcal{E}+s)$ on k gives $\theta s > 7/5000 + (1-\theta)E_8(k)/D$ for $0 < \theta \le 1$.

For an enlarged kernel removing k, let $\Pi = I - k\otimes k/D$, $B' = \Pi B$ and $C' = \Pi C$. Exact projection gives

$$B'^*B' = B^*B - (B^*k)(B^*k)^*/D, \quad C'^*C' = C^*C - (C^*k)(C^*k)^*/D.$$

Since $V_0^*k = 0$, zero sampling and zero/feature cross Grams remain unchanged. Injectivity of V_0^*C implies the projected rank is still 56. Domain preservation follows from S0. An action comparison obtained by subtracting $A_W C$ before projection retains a term $d\|CX\|$, as well as the new continuum error on $C'X$; it cannot replace $\|CX\|$ by $\|C'X\|$ without a separate estimate. These identities establish a permissible changed certificate, not a new numerical improvement or a barrier on that larger quotient.

S9 Admissible enrichment and a constructive action estimate

Let $Y = Bb + V_0 v : \mathbb{C}^{56} \to H$ use the supplied exact binary coefficient arrays. Each column is in $\mathrm{Dom}(G)\cap q^\perp$ by S0. A solve may choose the arrays but imposes no accuracy, rank or enlarged Gram-inverse premise. In particular the old frame-only error η is not available on Y. Put $\ell_8 = m_\Gamma - k_8$ and $L_8 = M_w \ell_8(D) M_w$. The pointwise identity $d_{8,+} = \ell_8 + r_8$ and S6 give, on the whole operator domain,

$$\mathcal{E} = SL_8S + R_m - 2p_Y\otimes p_Y + \Delta_R, \qquad \Delta_R = R_q - R_m, \quad \|\Delta_R\| \le \epsilon_R. \tag{S9.1}$$

This replaces the unpriced high-zero action on the enriched columns by a smooth multiplier action.

For one column with base representation $y_0 = w\sum_a c_a \sin(ax) + c_p w \sinh(x/2)$, write $y = Sy_0$ and $\alpha = \langle y_0, q\rangle/Q$. Then

$$\widehat{wy}(t) = i\left\{\sum_a c_a \frac{\Omega(t-a) - \Omega(t+a)}{2} + c_p v(t) + \alpha t \Xi(t)\right\}. \tag{S9.2}$$

The coefficients include the mathematical square roots of the actual masses. For V_0 columns the score overlaps vanish at the actual zero ordinates. The pole overlap is exactly $-1/4$.

Here is an explicit finite action enclosure. Put $a_0 = 1/4$ and $b_0 = 1/8$. For $|s| \le 1/4$, the complete theta series proves

$$|\Omega(t+is)|,\ |W\widehat{\sinh(x/2)}(t+is)|, |\Xi(t+is)| \le 20e^{-|t|/8}. \tag{S9.3}$$

For an explicit bound use $\pi\cos(1/4) > 5/2$ and $|\cosh((x+i\theta)/2)| \ge \frac{1}{2}\cosh(x/2)$. On $x \ge 0$, the weighted W majorant is

$$\sum_n [160n^4 e^{17x/4} + 76n^2 e^{9x/4}] e^{-(5/2)n^2 e^{2x}}.$$

Substitute $t = e^{2x}$ and bound its fractional powers by t^2, t. Their integrals are at most $106/125$ and $14/25$, times $e^{-5n^2/2}$. The lattice sum is at most $25/246$, so both half-lines cost $2[80(106/125) + 38(14/25)](25/246) < 20$. The other two functions have smaller bounds. Contour shifting by $1/8$ gives (S9.3), including both physical tails.

The analytic continuation $m_\Gamma(z) = [\psi(1/4+iz/2) + \psi(1/4-iz/2)]/2 - \log\pi$ has no poles in this strip. The complete trigamma series bounds its vertical derivative by 36, and the six-term cosine bound together with (S0.1) gives

$$|\ell_8(t+is)| \le 200 + 10|t|, \qquad |s| \le 1/4.$$

Regarding c_a, c_p, α as coefficient rows for the whole map Y, a common real and strip-line L^1 bound for $\ell_8(z)\widehat{wY}(z)$ is

$$A = 20\left[\sum_a (4480 + 160|a|)\|c_a\|_2 + 4480\|c_p\|_2 + 47200\|\alpha\|_2\right]. \tag{S9.4}$$

The constants come from the complete moments $\int e^{-|t|/8} = 16$, $\int |t|e^{-|t|/8} = 128$ and $\int t^2 e^{-|t|/8} = 2048$.

Let $h > 0$, $T = Kh$, $P = 2\pi/h$ and physical radius $R > 0$. If L_k denotes the exact row $\ell_8(kh)\widehat{wY}(kh)$ and $\widehat{L}_k$ its exact binary approximant, define

$$\widehat{g}(x) = \frac{h}{2\pi}\sum_{|k|\le K} \widehat{L}_k e^{-ikhx}, \quad \widehat{J} = S[w\widehat{g}],$$

$$t_h = \frac{h}{2\pi}\sum_{|k|>K}\|L_k\|_2, \quad n_h = \frac{h}{2\pi}\sum_{|k|\le K}\|L_k - \widehat{L}_k\|_2, \quad B_h = \frac{h}{2\pi}\sum_{k\in\mathbb{Z}}\|L_k\|_2.$$

Every infinite sum is bounded by shifted exponential envelopes in (S9.3), or the sharper real bounds (S0.5). Geometric series and their first two derivatives sum their polynomial factors. No primitive frequency is replaced by a smaller presumed band. Poisson summation and contour inversion give the complete map bound

$$\begin{aligned}\|SL_8Y - \widehat{J}\| \le \|w\|_2 &\left[\frac{A}{\pi}e^{a_0R}\frac{e^{-a_0P}}{1-e^{-a_0P}} + t_h + n_h\right] \\ &+ \left[\frac{A}{2\pi} + B_h\right]\|w1_{|x|>R}\|_2.\end{aligned} \tag{S9.5}$$

Indeed the inverse transform has row norm at most $Ae^{-a_0|x|}/(2\pi)$. Its periodic aliases on $[-R,R]$ sum to the first displayed term. Outside that interval use $A/(2\pi)$ and B_h on the two full inverse transforms, then multiply by w. Frequency omission and node error cost respectively t_h, n_h. The projection S is contractive. This proves a whole output-space map estimate, including components outside every trial span.

S10 The executed action certificate and its full Grams

The parameters are $h = 1/20$, $T = 3300$, $R = 3$. Near Fourier arguments use the complete Taylor enclosures in S4, with Horner, coefficient, zero-location and mass-square-root errors. The signed multiplier is evaluated at every exact grid point by outward digamma and six cosine balls. Score values through 128 use order-320 quadrature with the complete base error in S4; beyond 128 they use (S0.5). The following upward endpoints include all nodes and tails:

Quantity	Upper bound
Strip row integral A	$1.001591 \cdot 10^{9}$
Alias before $\|w\|_2$	$1.532843 \cdot 10^{-5}$
Finite node/location/rounding error before $\|w\|_2$	$1.797275 \cdot 10^{-8}$
Omitted frequencies before $\|w\|_2$	10^{-88}
Both physical tails after w	10^{-157}
Finite midpoint-action map error	$3.366659 \cdot 10^{-6}$
Complete map error g	$1.421821 \cdot 10^{-5}$
Trial norm $\|Y\|$	2.159290

Here $\|w\|_2 \le 1/\sqrt{2}$ and $\|w1_{|x|>3}\|_2 \le 2^{-550}$. The finite midpoint action is Bd, where d is the exact binary approximation to $G_B b + Q_b^* v$ with its pole row removed. Its coefficient error is priced as a physical map using $\|B\|$ and the full input Gram errors. Set $\widehat{A} = \widehat{J} + Bd$. Then

$$\|\mathcal{E}Y - \widehat{A} - \Delta_R Y\| \le g. \tag{S10.1}$$

The full physical action Gram is essential. For the smooth part it is

$$(h/(2\pi))^2 \widehat{L}^*[\Omega((k-l)h)]\widehat{L} - (h/(2\pi))^2 (b_0^*\widehat{L})^*(b_0^*\widehat{L})/Q.$$

The Toeplitz product uses native outward complex DFT and inverse at length 262144, on exact binary input. Every returned radius is retained. Kernel values through 128 use the complete Taylor certificate; the omitted kernel has its whole geometric L^1 price. The pole, score, Bd and all physical/Fourier cross products receive their full point and absolute-product guards. For $G_Y = Y^*Y$, $CY = C^*Y$, $YA = Y^*\widehat{A}$, $CA = C^*\widehat{A}$, $AG = \widehat{A}^*\widehat{A}$, the resulting operator uncertainties have upward endpoints

G_Y	0.014115739
CY	$4.979513 \cdot 10^{-5}$
YA	$5.596759 \cdot 10^{-6}$
CA	$1.982337 \cdot 10^{-8}$
AG	$2.263431 \cdot 10^{-9}$

These are matrix uncertainties; they are not silently treated as a whole-map error. In particular AG includes the entire outside output.

S11 Shifted residual theorem and the finite tradeoffs

Put $T_s = \mathcal{E} + sI$, $s = \mu - \rho > 0$. Completing the square gives

$$C^* T_s^{-1} C = K + R^* T_s^{-1} R \le K + R^* R/s, \quad R = C - T_s Y, \quad K = C^* Y + Y^* C - Y^* T_s Y.$$

This uses $T_s \ge sI$ and the operator-domain membership of all trial columns, without an unshifted gap. Define

$$K_s = CY + CY^* - sG_Y - (YA + YA^*)/2,$$
$$S_s = H_C - CA - CA^* - s(CY + CY^*) + AG + s(YA + YA^*) + s^2 G_Y.$$

Write the exact error as $\Delta_R Y + D$, $\|D\| \le g$. Its quadratic contribution is at most $\epsilon_R G_Y + g\|Y\| I$, and for $\tau > 0$ its squared Gram is at most $(1+\tau)\epsilon_R^2 G_Y + (1+1/\tau) g^2 I$. A second Young inequality separating this error from the whole residual $C - \hat{A} - sY$ gives, for every $\beta > 0$,

$$C^* T_s^{-1} C \le K_s + \left[\epsilon_R + \frac{(1+1/\beta)(1+\tau)\epsilon_R^2}{s}\right] G_Y + \frac{1+\beta}{s} S_s + \left[g\|Y\|_{\mathrm{up}} + \frac{(1+1/\beta)(1+1/\tau) g^2}{s}\right] I. \tag{S11.1}$$

The chosen $\tau = g/(\epsilon_R \|Y\|_{\mathrm{up}})$ is positive. Each primitive matrix uncertainty from S10 is multiplied by its actual absolute scalar coefficient in the assembly of K_s, S_s. Exact shift conversion and all scalar products receive their stated rounding guards. The finite spectral comparison uses (S2.2) with exact-input 192-bit Arb residuals. Multiplication of the entire right side by 11/10 and division of its upper endpoint by $n_0 = 1 - 10^{-8}$ yield an upward rational θ on the 10^{-12} lattice. By (S3.1) and the factorization norm identity,

$$R_{56} \le \theta T_s, \qquad \mathfrak{a}(u) \ge (1-\theta) E_8(z) - [\rho + \theta s]\|z\|^2, \quad z = (I - P_K)u.$$

The final transfer is exactly (S0.2); it does not recompute a numerical certificate on a changed frame.

There are 231 tested rational parameter pairs with individually enclosed endpoints. Some have $\theta \ge 1$ and do not retain a strictly positive energy coefficient. Among those with $\theta < 1$, the least scalar loss is the first row below. The stated theorem uses exact terminating coefficients; scalar losses in the table are rounded upward for presentation. The exact first loss is given below the table.

μ	β	$1 - \theta$	scalar loss upper
401/12800	1913/2048	0.000118321042	0.031324476155738683
7/200	1	0.065210231692	0.032749571
1/25	3/2	0.132584532412	0.034762
9/200	9/4	0.182372427581	0.036882534
1/20	1277/512	0.212458845389	0.039482
3/50	1	0.236184893938	0.045944547
15867/102400	63/512	0.294954254754	0.109392080

The first exact loss is $313244761555738682727829/10^{24}$. Every successful final price is strictly below θn_0. The set consists of a finite grid and rational local refinements; no untested interval is excluded. For fixed positive parameters the central bound has form $A + \beta s G_Y + M/s$ with $M \ge 0$, so it is operator convex in s, but its derivative has no fixed sign. It therefore supplies no monotone bisection or global optimality result.

S12 Exact signed-action cancellation

The full identities in S6 and (S9.1) give

$$A_Y = \mathcal{E} - R_q - T = SL_8S - 2p_Y \otimes p_Y - T, \qquad T = S(J - J_8)S, \quad \|T\| \le t_8.$$

Let $\widehat{J}$ be the actual smooth Fourier action from S9–S10, and define $\widehat{Z} = \widehat{J} - 2p_Y \otimes (Y^* p_Y)$ with its exact physical pole pairing. The continuum term has canceled before approximation. Thus $A_Y Y - \widehat{Z} = (SL_8 Y - \widehat{J}) - TY$ and the rational whole-map bound is

$$\begin{aligned}\|A_Y Y - \widehat{Z}\| \le \frac{71}{100}\left(\frac{1532843}{10^{11}} + \frac{1797275}{10^{14}} + 10^{-88}\right) + 10^{-157} \\ + \frac{32237}{10^{12}}\frac{2159290}{10^{6}} < \frac{11}{10^{6}}.\end{aligned}$$

Every prime tail, Fourier tail and physical output remains. This improves a signed-action approximation cost, not the global form floor. The combined positive-action cache by itself does not determine all separate smooth/pole cross Grams for this approximant. On a larger verified quotient, both trial and approximant must be projected. Contractivity preserves the error but not unchanged rank or inverse claims.

S13 A cached positive five-dimensional subspace

A different signed approximant can be formed from the existing combined cache. Write $U = \widehat{A}$ and choose the supplied exact binary selector X for $N^{-1}C^*Y$. Set $Z_s = U - CX$. The full dense capture gives $0 \le R_m - CN^{-1}C^* \le (3/12500)I$, hence

$$\begin{aligned}\|A_Y Y - Z_s\| \le g + (3/12500 + t_8)\|Y\| \\ + \|C\|\,\|N^{-1}C^*Y - X\| < 7/12500.\end{aligned}$$

The selector error is paid by its exact-input solve residual, $N \ge n_0 I$, the full CY uncertainty, and the whole saved-center N error at most $2 \cdot 10^{-8}$. Its whole physical Grams are

$$Y^*Z_s = YA - CY^*X, \quad C^*Z_s = CA - H_C X,$$

$$Z_s^*Z_s = AG - CA^*X - X^*CA + X^*H_C X.$$

All input and saved binary-center errors are included; in particular AG is not replaced by a projected action Gram.

Let W_5 be the supplied exact 56×5 selector and $V = YW_5$. The enclosed physical Gram is above $0.5046238284239I$ and $\|V\|^2 < 5$. The Hermitian signed center, after subtracting its finite error and $(7/12500)\|Y\|\|W_5\|^2$, exceeds $0.0012532198376I > I/1000$. Therefore

$$V^*V > \tfrac{1}{2}I, \qquad \mathfrak{a}(f) > \|f\|^2/5000 \quad (0 \ne f \in \operatorname{ran} V). \tag{S13.1}$$

The input norm here is Euclidean until the displayed physical-norm conversion using $\|V\|^2 < 5$.

Let P_5 be the actual projection onto $\operatorname{ran} V$. A supplied exact selector L approximates the projection of Z_sW_5 into V. Its full residual Gram is $(Z_sW_5 - VL)^*(Z_sW_5 - VL)$, with every mixed term retained. The certified lower bound for V^*V justifies its inverse and prices the difference between the selector and the exact projection. Subtract that price and $(7/12500)\|W_5\|$ from the certified largest residual norm for a lower bound. For an upper squared norm, Young's inequality with

parameter 1/100 retains both the residual and whole-map error. The exact-input residual/polar enclosures give

$$\|(I-P_5)A_Y V\| > 11/50, \qquad \|(I-P_5)A_Y V\|^2 < 0.062842088203. \tag{S13.2}$$

These are norms of the map with the ordinary Euclidean coefficient input of V, not a normalized physical metric. A large coupling is not a negative signed direction. Nor is $\operatorname{ran} C$ proved to lie inside this five-dimensional space, so its complement does not inherit the $-\rho$ floor.

S14 A fully coupled Schur criterion and its scope

Keep the old frame by taking $J_* = [C, Y]$ and $F = \operatorname{ran} J_*$. Every column lies in $\mathrm{Dom}(A_Y)$, and (S2.1) implies $\mathfrak{a}|_{F^\perp} \geq -\rho I$ because $C^* v = 0$ there. This statement requires neither an inverse of $J_*^* J_*$ nor a certified rank of the union. Let Z_* approximate $A_Y J_*$ and suppose $D = A_Y J_* - Z_*$ satisfies $D^* D \leq E$, with $E \geq 0$. For any exact coefficient selector L, put

$$H = J_*^* J_*, \quad B = \mathrm{Herm}(J_*^* Z_*), \quad Q_L = (Z_* - J_* L)^*(Z_* - J_* L).$$

For $d > \rho$ and $\beta, \eta > 0$, the sufficient comparison is

$$B + (d - \eta/2)H - \frac{E}{2\eta} - \frac{(1+\beta)Q_L + (1+1/\beta)E}{d-\rho} \geq 0. \tag{S14.1}$$

It proves $\mathfrak{a}(u) \geq -d\|u\|^2$ on the full score quotient. Indeed write $u = J_* x + v$, $v \perp F$. Young's inequality gives $\mathrm{Herm}(J_*^* D) \geq -(\eta H + E/\eta)/2$. For the actual projection P_F, $(I - P_F)Z_* = (I - P_F)(Z_* - J_* L)$, so the outside true-action Gram is at most $(1+\beta)Q_L + (1+1/\beta)E$. Complete the square against the complementary lower bound $(d-\rho)\|v\|^2$. This proves (S14.1) even when the representation by J_* is nonunique. It is the standard Schur principle with the required coercivity explicitly supplied, not a free unshifted inverse [5, Theorem 1.2]. For block map errors e_C, e_Y, one may use $E = \mathrm{diag}((1+\lambda)e_C^2 I, (1+1/\lambda)e_Y^2 I)$. The mixed signed-action Grams needed to verify this criterion for $[C, Y]$ have not been established in the present release.

There is also a precise limitation for one declared Y-only capture class. If $W \geq 0$ and $YWY^* \leq I$, then $YWY^* \leq P_Y$. Consequently a sufficient complement price inherited from (S2.1) is

$$\rho_{\mathrm{eff}} = \rho + \frac{11}{10 n_0}\|H_C - CY\, W\, CY^*\|.$$

Let G_c, CY_c be the saved centers and $g_0 = 0.014115739$ exceed the accepted G_Y error. The positive matrix $M = G_c + g_0 I$ is invertible, and $W_g = M^{-1}$ is the largest capture in the cone $0 \leq W \leq W_g$. Every smaller capture leaves a larger positive residual. An exact binary Rayleigh locator for the center residual gives the outward value $0.13436493992529662392491776687409$. With $h = 32/10^{10}$, $c = 1/20000$, its full perturbation is at most

$$h + 2c\|W_g CY_c^*\|_F + c^2\|W_g\|_F < 0.000384941947.$$

The native 192-bit inverse/Rayleigh check therefore yields $\rho_{\mathrm{eff}} > 0.1478676174 > 0.14$ throughout that cone. This is a bound on that sufficient certificate's price, not a lower bound on a negative eigenvalue of the actual signed complement.

Certificate correspondence and verification boundary

The section numbers above are the precise supplementary references in the main article. The manifest names the immutable binary inputs, executed-source snapshots and certificate records for each calculation. The correspondence is as follows:

S0, S4	symbol cells, Fourier grids, physical/zero tails
S1–S3	midpoint factors, dense residual, physical norms and Gram
S4	low-zero roots, Taylor grid and full sample/correlation matrices
S5	lower masses and weighted deficit norm
S6–S7	energy resolvent and whole matrix residual
S8	equality-jet integrals and fixed-price barrier
S9–S10	enriched trial, whole Fourier action and full Grams
S11	rational shift endpoints and their selected matrices
S12–S13	signed cancellation, cached subspace and outside coupling
S14	inverse/Rayleigh robust-capture check

The bounded release verifier checks hashes, exact rational consequences, selected enriched comparisons reconstructed from cached matrices, and the stated robust-capture inverse/Rayleigh calculation. It accepts the upstream quadrature, zero completeness, Taylor, large-Gram and whole DFT/action allowances as certificate inputs. It does not regenerate those expensive computations. The analytic arguments in S0–S14 explain how those inputs control the actual continuum problem; a successful finite replay is not by itself a formalization of those arguments. The release verification report and formal correspondence document list the checked Lean declarations and their explicit analytical inputs.

References

[1] L. van Hemert, *Sharp logarithmic coupling estimates for theta equality frames in a signed Weil-form realization*, v1.0.0, 11 October 2026, doi:10.5281/zenodo.23298499, Lemmas 2.1–2.2 and 5.1.

[2] NIST Digital Library of Mathematical Functions, equation 5.9.13.

[3] E. Hasanalizade, Q. Shen and P.-J. Wong, *Counting zeros of the Riemann zeta function*, J. Number Theory 235 (2022), 219–241, Corollary 1.2, equation (1.5), author-hosted published article.

[4] FLINT project, *FLINT 3.6.0*, versioned `acb_dirichlet` documentation.

[5] G. Dusson, I. M. Sigal and B. Stamm, *The Feshbach–Schur map and perturbation theory*, arXiv:2105.02058v1, Theorem 1.2.